

STEPPER MOTOR CONTROLLER USING 'C'

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Stepper motors are widely used in robots and other automation applications where precise positioning is required. A stepper motor, unlike ordinary motors, runs in a

number of single steps to complete the required rotations.

Here is a simple program (page 60) written in 'C' language for controlling a 4-lead stepper motor, used in TVS dot-matrix printers, through the parallel port of a PC. The program can control the number of rotations, direction of rotation (clockwise or anticlockwise) and rotational speed of the motor.

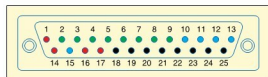


Fig. 1: 25-pin parallel-port connector

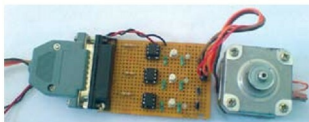


Fig. 2: Author's prototype of the stepper motor controller

Pin Details of 25-pin, D-type Parallel-Port Connector

Pin	Register	Signal bit	Signal direction	Level	Hardware
1	Control	C 0	Output	Low	Inverted
2	Data	D 0	Output	High	Direct
3	Data	D 1	Output	High	Direct
4	Data	D 2	Output	High	Direct
5	Data	D 3	Output	High	Direct
6	Data	D 4	Output	High	Direct
7	Data	D 5	Output	High	Direct
8	Data	D 6	Output	High	Direct
9	Data	D 7	Output	High	Direct
10	Status	S 6	Input	High	Direct
11	Status	S 7	Input	Low	Inverted
12	Status	S 5	Input	High	Direct
13	Status	S 4	Input	High	Direct
14	Control	C 1	Output	Low	Inverted
15	Status	S 3	Input	High	Direct
16	Control	C 2	Output	High	Direct
17	Control	C 3	Output	Low	Inverted
18-25	Ground				

Note: Control register can be used as output as well as input

Basics of the parallel port

The parallel port of the PC is a 25-pin, D-type connector, commonly known as LPT (line printer terminal) port. It has three different ports, namely, data port, control port and status port. The parallel port can be accessed via 25-pin, D-type female connector (DB25) as shown in Fig. 1. This female connector is normally available at the back side of your PC.

In order to access any port, you should have its port ID, i.e., the base address of the port. Here, the base address of the data port is '0378' in hex format. The addresses of the three ports are in sequential order. So if the address of the data port is '0x378', the address of the status port and the control port will be '0x378+1'

and '0x378+2' respectively.

In DOS, you can use the debug.exe program to check the port address of your PC. DOS Debug is a byte editor that allows files to be viewed and modified at the byte level. It is a standard feature of many modern DOS versions, including MD-DOS version 7. While MS-DOS is not commonly used today, it can still be accessed from Windows 95 and Windows 98 by clicking 'Start'/'Run' buttons followed by typing 'command' or 'cmd' in Windows NT/2000/XP.

Run the debug program by typing 'debug' against the DOS prompt and then type 'd 40:0008' against the '-' prompt (debug prompt). You will get something like '78 03 00 00 00 00 00 00.' Here, '78 03' stands for the port address in reverse order of '0378' in hexadecimal. Press 'Q' to exit the debug program.

Circuit for interfacing the parallel port

Not all the 25 pins of the port are used for every application. Normally, the eight data lines (pins 2 through 9) along with the signal ground line are used.

Pins 2 through 9 form the data port, thus making it an 8-bit wide port. It is a unidirectional port. That is, it can be used only to output the data to the connected device. Pins 1, 14, 16 and 17 form the control port. It is a bidirectional port, i.e., it can be used both to input and output the data to and from a device. Pins 15, 13, 12, 10 and 11 form the status port. It is a unidirectional port, i.e., it can be used only for data input from a device. The various pin functions of the parallel port are shown in the table.

